

Chapter 10 Objectives: Be able to...

- Explain the interrelationship between the **light reactions** and the Calvin cycle of photosynthesis and list the **inputs** and **outputs** of each.
- Explain how **electrons** are excited by **light energy** and describe the **4 potential fates of excited electrons**.
- Arrange the components of the photosynthetic machinery within the **chloroplast structure**.
- Explain how light energy is absorbed by the **photosynthetic antenna complex**.
- Trace the path of an e^- through the **e^- transport chain**, and explain how this generates **electrochemical potential, ATP, and reducing potential**.
- Describe the selective adaptations of **C4 and CAM photosynthesis**.
- Compare **e^- flow** between **oxidative phosphorylation** and **photophosphorylation**.

Redox Reactions Involve the Gain or Loss of Electrons

Cellular respiration *releases* energy from sugar

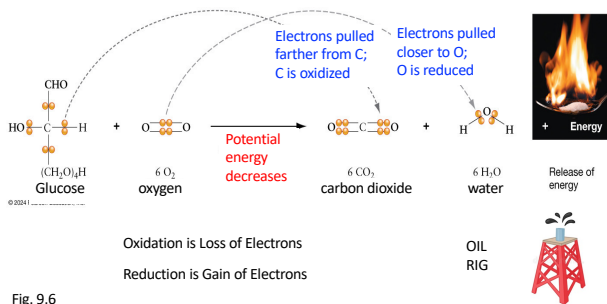


Fig. 9.6

Redox Reactions Involve the Gain or Loss of Electrons

Photosynthesis *stores* energy in sugar

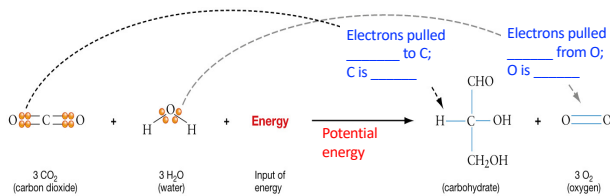
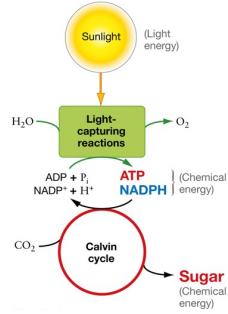


Fig. 10.1

Photosynthesis: Light Reactions + Calvin Cycle



Light Reactions:

Calvin Cycle:

Fig. 10.2

Chloroplasts: Multiple Compartments

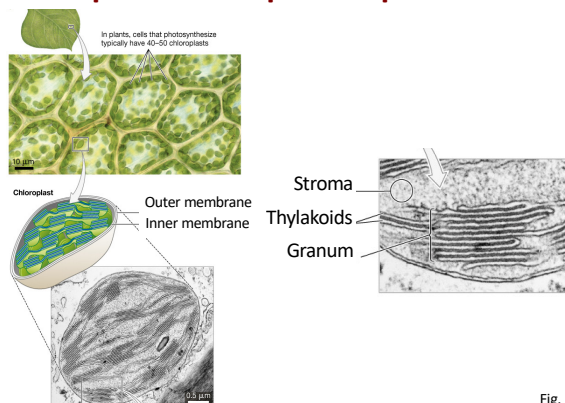


Fig. 10.3

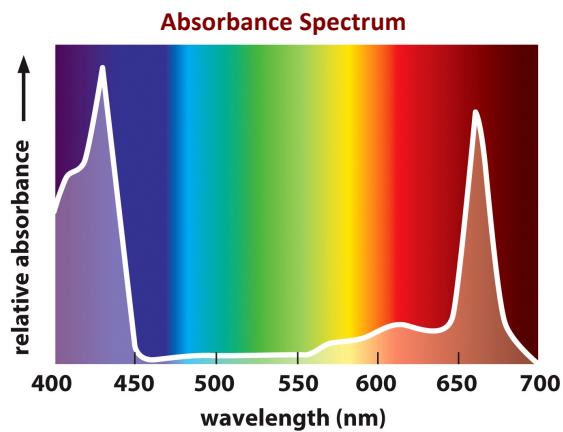


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The diagram illustrates the Fraunhofer experiment. At the bottom, a light source emits a beam of white light, which passes through a slit and is dispersed by a prism into a spectrum of colors (violet, blue, green, yellow, orange, red). Above the spectrum is a horizontal tube containing a culture. The tube is divided into sections corresponding to different wavelengths of light. The top of the tube is labeled 'Wavelength (nm)' with markers at 400, 500, 600, and 700. The bottom of the tube shows the distribution of organisms: 'Aerotactic bacteria' (represented by black dots) are concentrated in the blue and violet regions (shorter wavelengths), while 'Spirogyra cell' (represented by green wavy lines) are concentrated in the red and orange regions (longer wavelengths). A 'Spiral chloroplast' is also labeled within the Spirogyra cell.

(a) Chlorophyll a

CH₃ in chlorophyll *a*
CHO in chlorophyll *b*

Head
(ring structure that absorbs light)

Hydrophobic Tail
(anchors chlorophyll in thylakoid membrane)

(b) β-Carotene

[illegible]

relative absorbance

400 450 500 550 600 650 700

wavelength (nm)

Shorter wavelength = higher energy

2

1

0

Photon

Photon

e^-

Blue photons excite electrons to an even higher energy state

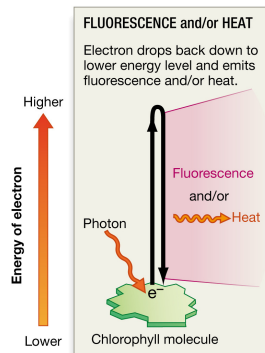
Red photons excite electrons to a high-energy state

Ground state

Possible energy states of electrons in chlorophyll

Fig. 10.9

Four Fates for Excited Electrons in Photosynthetic Pigments



Energy is released as:

Fig. 10.10

Four Fates for Excited Electrons in Photosynthetic Pigments

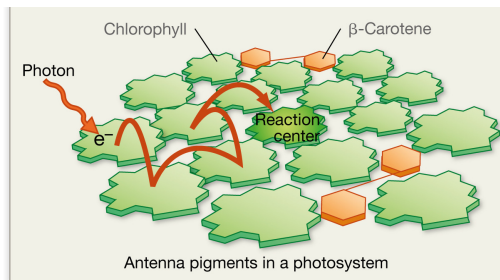


Fig. 10.10

Four Fates for Excited Electrons in Photosynthetic Pigments

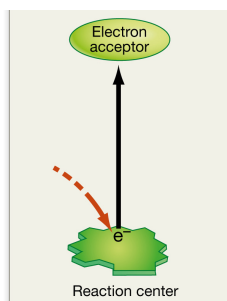
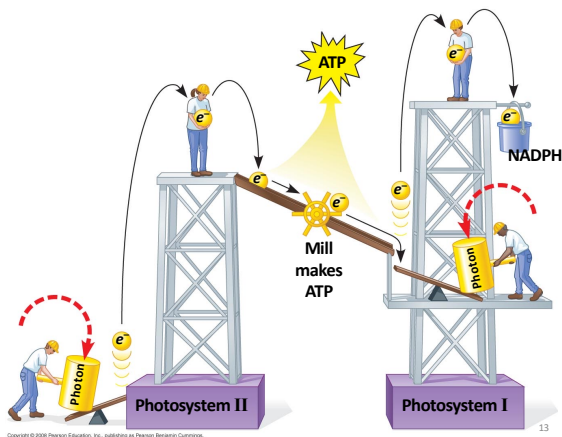


Fig. 10.10



Photosystem II Feeds Excited Electrons to an Electron Transport Chain

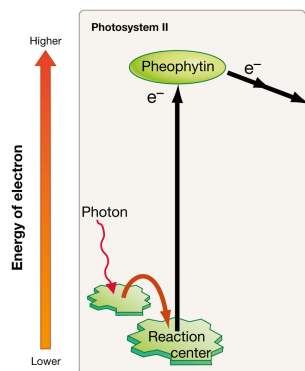


Fig. 10.12

Electron Transport between Photosystem II

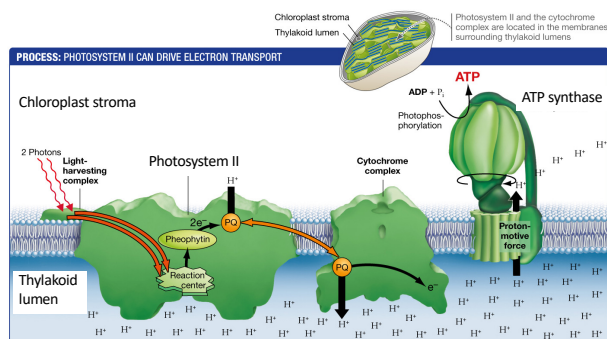


Fig. 10.13

Photosystem I Produces NADPH

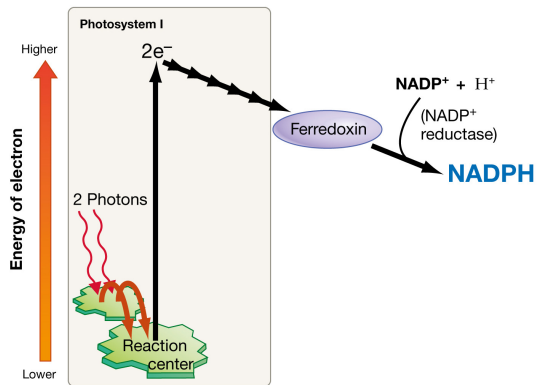


Fig. 10.14

Photosystems II and I are Linked “Z Scheme”

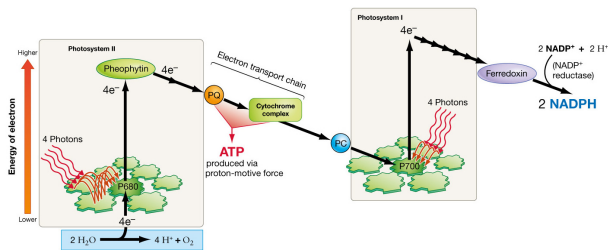


Fig. 10.15

Photosystems II and I are Linked

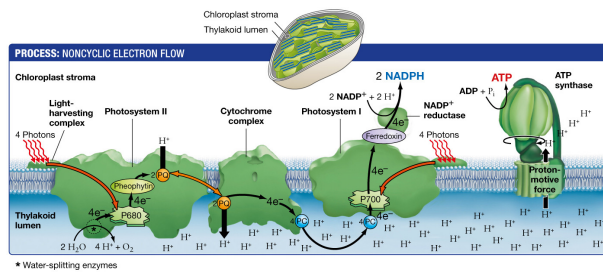
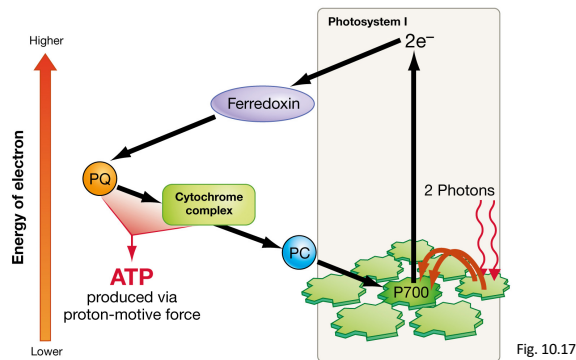
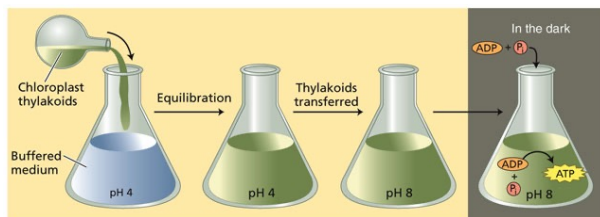


Fig. 10.16

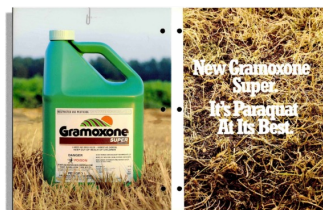
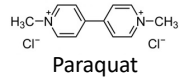
Cyclic Electron Flow



Chemiosmotic mechanism



Photosynthesis Inhibitors



Paraquat Lawsuit
What is Paraquat?

Paraquat is a highly toxic herbicide.

- EPA classified as "Restricted Use Pesticide" (RUP)
 - ingesting small amounts can still be fatal
 - only trained, certified applicators can use
 - illegal to transfer liquid to drink containers
- Commercial production began in 1961
 - One of the most commonly used herbicides

Paraquat is used to:

- Control weeds/grasses
- Support regenerative agricultural practices
- Aid soil conservation
- Reduce labor
- Harvest aid

Used on:

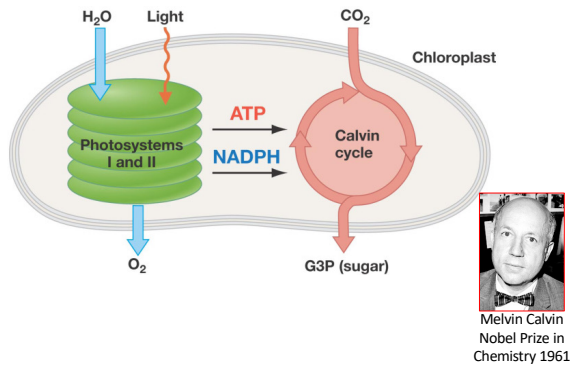
- Cotton
- Wheat
- Corn
- Almonds
- Strawberries
- Soy

Paraquat 200SL

DANGER

KING LAW

ATP and NADPH from light reactions



Leaf Cells Obtain Carbon Dioxide through Stomata

(b) Carbon dioxide diffuses into leaves through stomata.

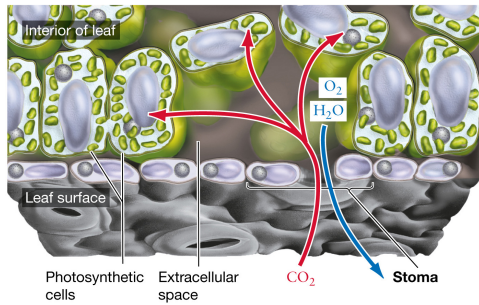


Fig. 10.19

Rubisco Is a Large Enzyme Complex

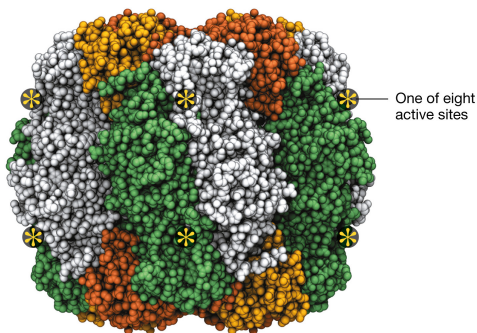
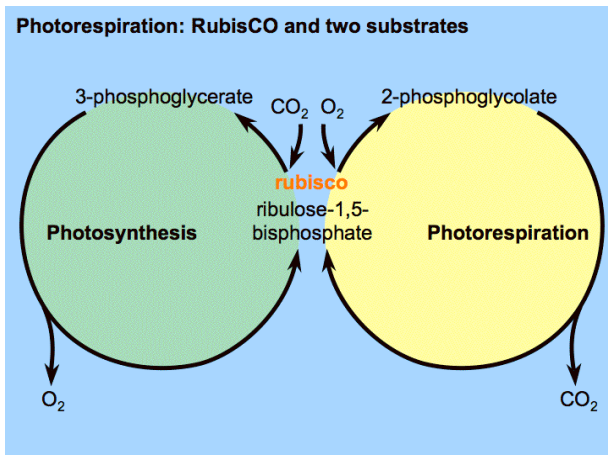
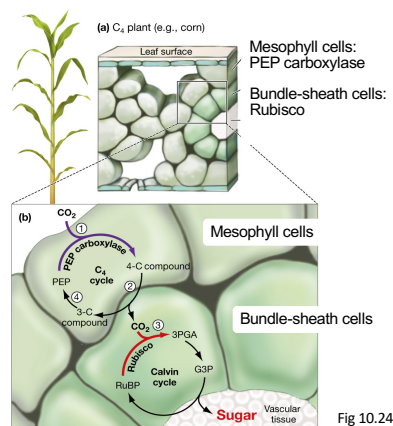


Fig. 10.21

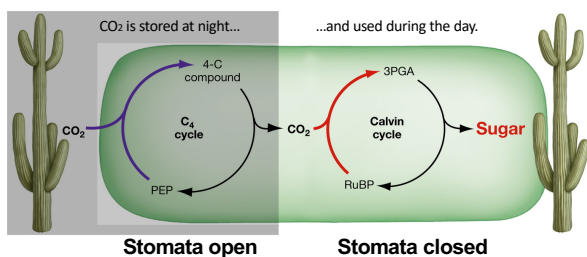
Photorespiration: RubisCO and two substrates



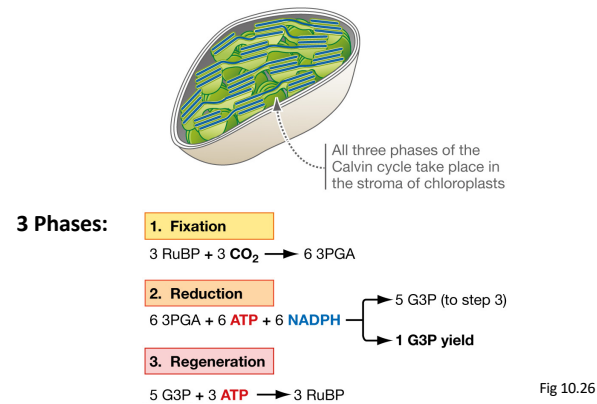
C4 photosynthesis:



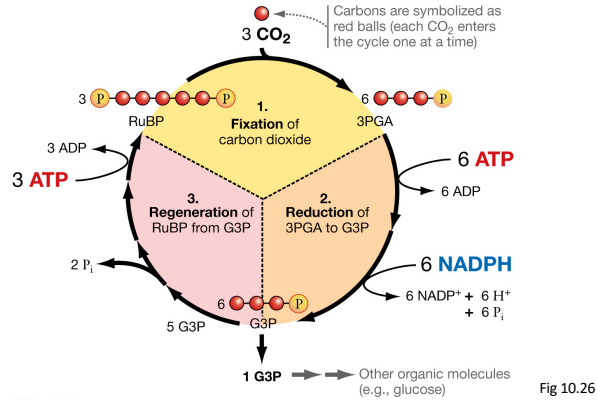
Crassulacean acid metabolism (CAM):



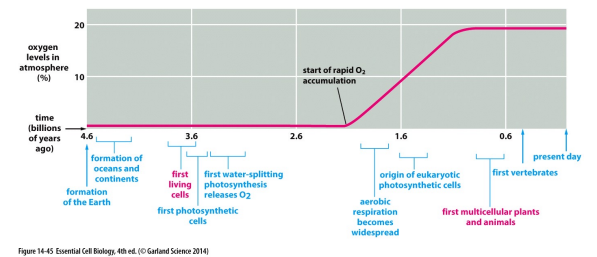
The Calvin Cycle Fixes and Reduces the C in CO₂



The Calvin Cycle Fixes and Reduces the C in CO₂



Oxygenic photosynthesis evolved ~3bya



Plants use oxidative phosphorylation *and* photophosphorylation!

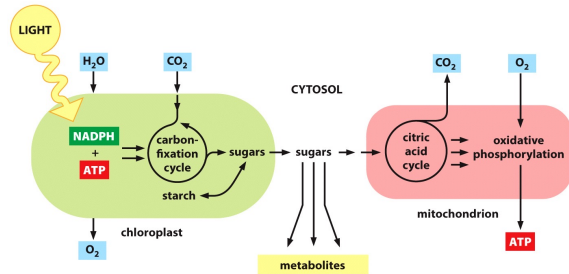
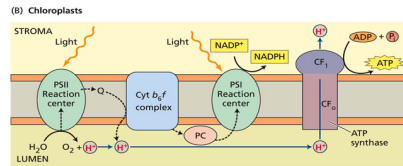


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Photophosphorylation vs. Oxidative phosphorylation

Photophosphorylation

- e⁻ donor is H₂O
- Ultimate e⁻ acceptor is NADP⁺



Oxidative phosphorylation

- e⁻ donor is NADH
- Ultimate e⁻ acceptor is O₂

